

Fundamentals of Data Science — CO3

Assessment Tool 2

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Data Interpretation: Questions 16–20 with Solutions

16. Continuous Distribution — Area Interpretation

Given: $\mu = 100$, $\sigma = 15$

(a) $P(85 < X < 130)$:

$$85 = \mu - \sigma \rightarrow z = -1$$

$$130 = \mu + 2\sigma \rightarrow z = +2$$

$P(z < 2) - P(z < -1) = 0.9772 - 0.1587 = 0.8185 \rightarrow$ approximately 81.85% of values lie between 85 and 130.

(b) $X = 145$:

$$z = (145 - 100) / 15 = 3.0 \rightarrow 3 \text{ standard deviations above the mean.}$$

Under the empirical (68–95–99.7) rule, only about 0.3% of values fall beyond $\pm 3\sigma$ (roughly 0.15% in each tail). A value this far out is rare enough to flag as a potential outlier.

17. Sampling Bias vs Sampling Error

(a) This is sampling bias, not sampling error. Sampling error is random fluctuation around the true mean that shrinks as sample size grows; here the discrepancy (₹18,000 vs ₹14,500) is systematic — online-panel respondents likely skew toward higher-income, more tech-connected households, which is a selection problem rather than chance variation.

(b) No. Increasing sample size only reduces the width of the sampling distribution (reduces error); it does not fix a biased sampling frame — it would simply produce a more precisely wrong estimate. Fixing bias requires a representative sampling method (e.g., stratified or random sampling across the full population), not a larger sample.

18. Covariance Sign vs Correlation Strength

$\text{Cov}(\text{Pair 1}) = 850$ vs $\text{Cov}(\text{Pair 2}) = 12$ — one cannot conclude Pair 1 has a stronger relationship, because covariance is scale-dependent: its magnitude depends on the units and variance of X and Y , not purely on how tightly they co-move. A pair measured in large units (e.g., rupees, thousands) will naturally produce large covariance values even with a weak relationship.

Missing information: the standard deviations (SD_x , SD_y) of each pair are needed to compute correlation, $r = \text{Cov}(X,Y) / (SD_x \cdot SD_y)$, which standardizes covariance to a unit-free scale (-1 to $+1$) for a fair comparison of relationship strength.

19. Estimation — Confidence Interval Reasoning

Given: $n = 100$, mean = 250, $SD = 40$

$$SE = SD / \sqrt{n} = 40 / \sqrt{100} = 40 / 10 = 4$$

$$95\% \text{ CI} = \text{mean} \pm 1.96 \times SE = 250 \pm 1.96(4) = 250 \pm 7.84$$

$$95\% \text{ CI} = (242.16, 257.84)$$

Interpretation

What it DOES tell you: if this sampling process were repeated many times and an interval built the same way each time, about 95% of those intervals would contain the true population mean. It is a statement about the reliability of the method, not a probability about a single fixed value.

What it does NOT tell you: it does not mean there is a 95% probability the true mean lies in this specific interval (the true mean is fixed, not random — it either is or isn't in the interval); it also does not describe the spread of individual data points, nor guarantee this particular sample is representative.

20. Integrated Case Study

Data (₹'000), $n = 15$: 120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

Sorted: 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 128, 129, 130, 500

(a) Outlier Identification

IQR method:

$Q1$ (median of lower 7) = 120; $Q3$ (median of upper 7) = 128

$IQR = 128 - 120 = 8$

Lower fence = $120 - 1.5(8) = 108$; Upper fence = $128 + 1.5(8) = 140$

$500 > 140 \rightarrow$ confirmed outlier

Z-score method:

Mean = $2227/15 = 148.47$; $SD \approx 94.0$

z for 500 = $(500 - 148.47) / 94.0 \approx 3.7$ (well beyond the typical ± 3 cutoff) $\rightarrow$ confirmed outlier

Both methods agree: 500 is a clear outlier.

(b) Mean and Median — With vs Without Outlier

	With outlier ($n=15$)	Without outlier ($n=14$)
Mean	148.47	123.36
Median	124	123.5

The mean shifts by about 25 units (~20%) when the outlier is removed, while the median barely moves (0.5 units). This confirms the mean is highly sensitive to the outlier, while the median is robust — making the median the more trustworthy central measure here.

(c) Skewness (With Outlier Included)

$Sk = 3(\text{Mean} - \text{Median}) / SD = 3(148.47 - 124) / 94.0 \approx +0.78$

A positive coefficient around 0.78 indicates a moderately-to-strongly right-skewed (positively skewed) distribution, consistent with a long right tail caused by the single extreme value (500).

(d) Recommendation

Investigate before deciding — do not blindly remove or blindly keep the value 500.

Since 14 of the 15 values sit tightly between 117 and 130, a jump to 500 is more consistent with a data-entry error (e.g., an extra digit or duplicated/misplaced value) than a natural spike — the source invoice or record for that day should be checked.

If confirmed to be an error: correct it if the true value is known, otherwise treat it as missing and impute or exclude it.

If it is a genuine event (e.g., a real bulk order or promotion): retain it but report it separately with a footnote, since removing genuine business data would hide real performance.

Either way, for monthly reporting, the median (or a trimmed mean) should be favored over the raw mean as the headline statistic — parts (b) and (c) show that both the mean and skewness are being distorted by this single point, so leading with the mean alone would misrepresent typical daily performance.